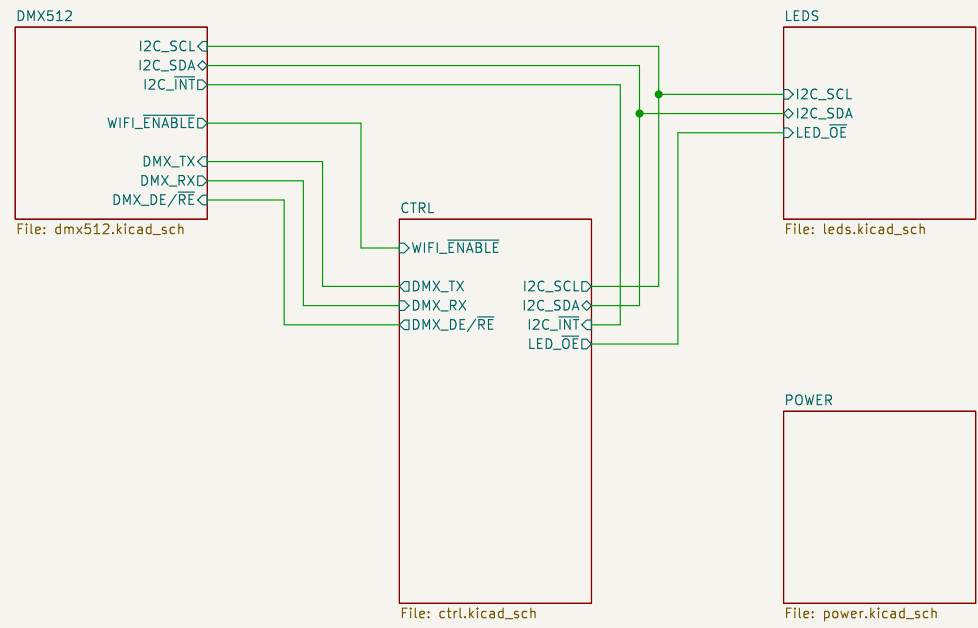


LED Beacon

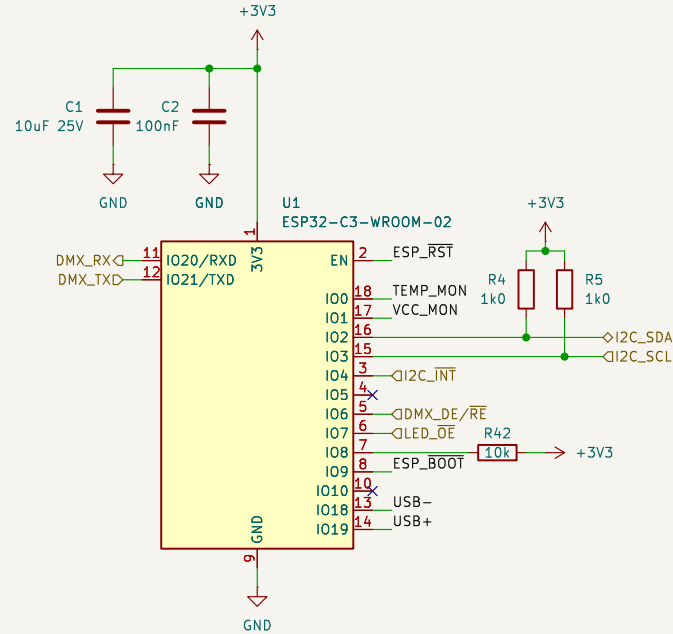
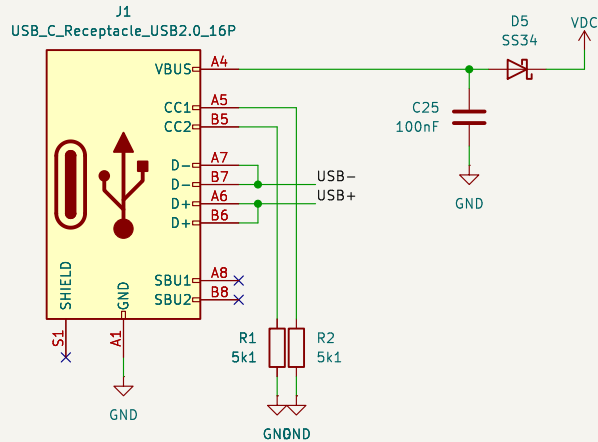


- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

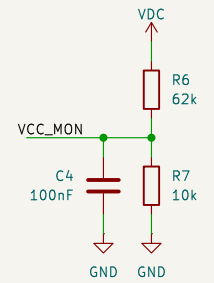
Wires & Bits		
Sheet: /		
File: emf_beacon.kicad_sch		
Title:		
Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 1/5

Controller

USB Programming

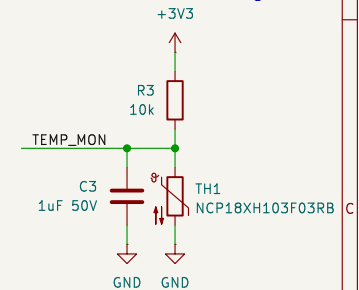


VCC Monitoring

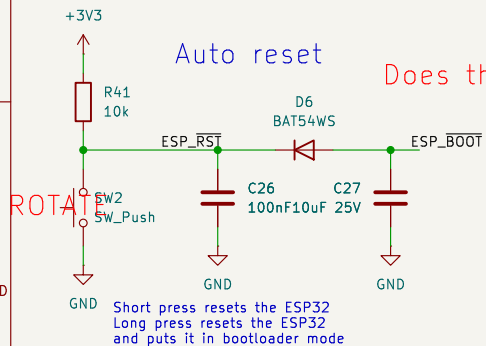


Voltage Monitor
 V_{max} is 17.5V
 $V_{adc} \times ((43k\Omega + 10k\Omega) + 10k\Omega) = VCC$
 $V_{adc} \times 5.3 = VCC$
 $1lsb = 4.271\mu V$

Temperature Monitoring



Auto reset

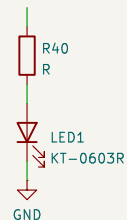


Short press resets the ESP32
 Long press resets the ESP32
 and puts it in bootloader mode

Does this need to be a switch?
 Cheaper switch?
 Side actuation?

Status LED

Status LEDs?



Wires & Bits

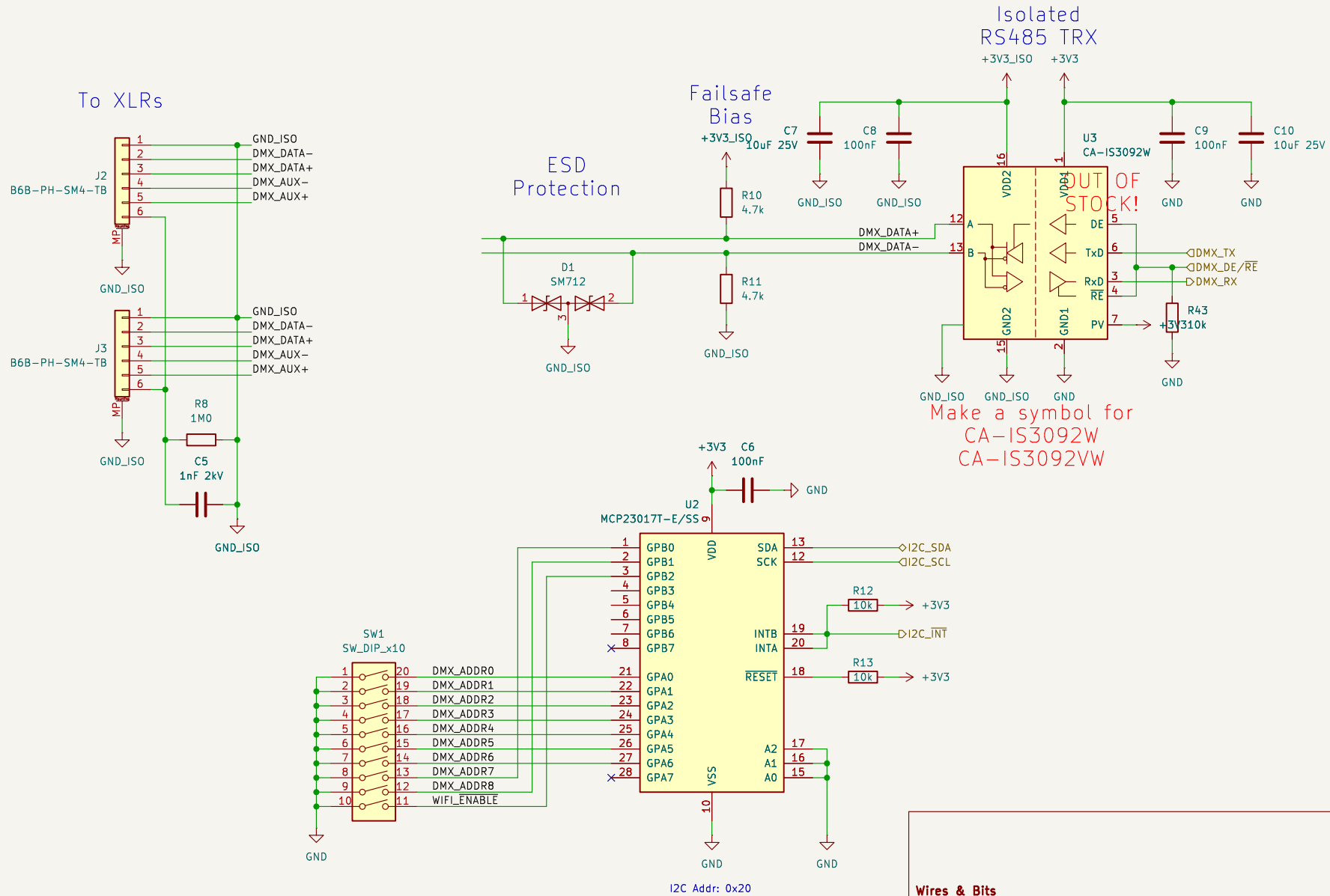
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 File: ctrl.kicad_sch

Title:

Size: A4
 Date: 2026-04-10
 KiCad E.D.A. 9.0.9

Rev: 0
 Id: 2/5

DMX512



Wires & Bits

Sheet: /DMX512/

File: dmx512.kicad_sch

Title:

Size: A4

Date: 2026-04-10

Rev: 0

KiCad E.D.A. 9.0.9

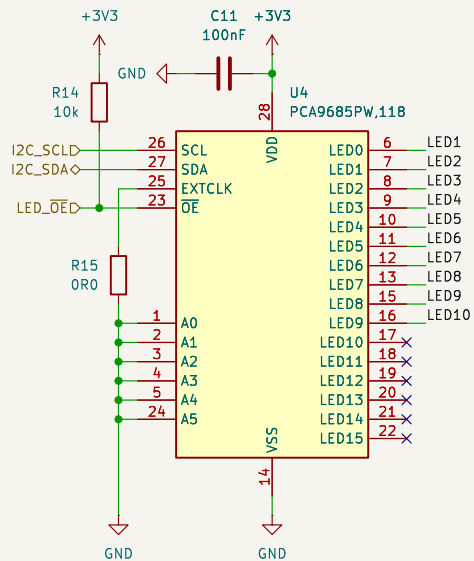
Id: 3/5

LED Driver

LED Enable

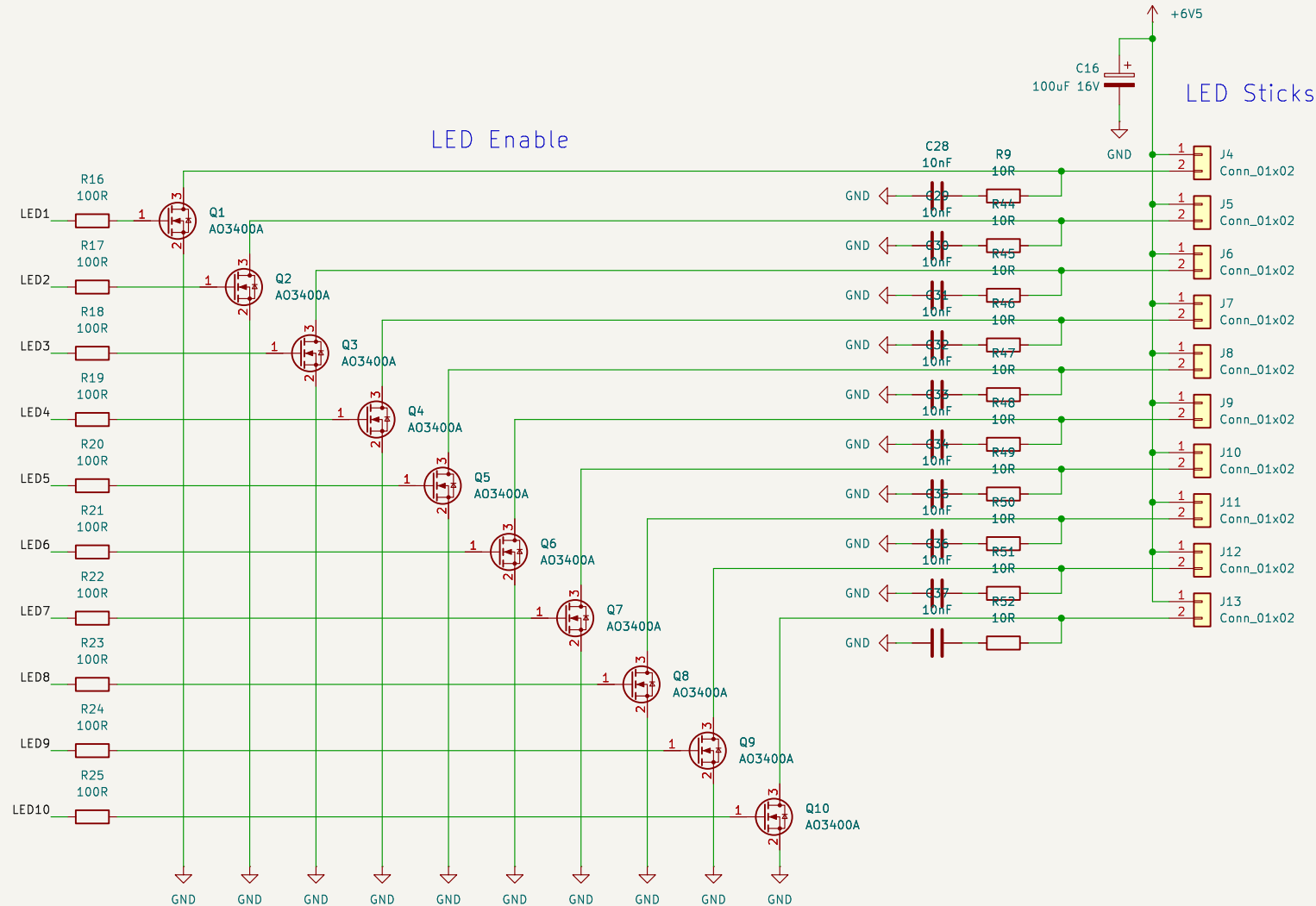
LED Sticks

I2C → PWM



I2C Addr: 0x40

A string of 6x LEDs is 6.4Vf @ 0.3A



Wires & Bits

Sheet: /LEDS/
File: leds.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.9

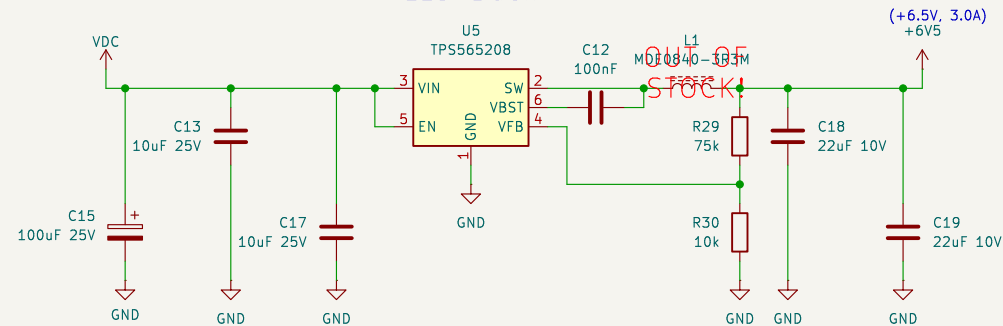
Date: 2026-04-10

Rev: 0

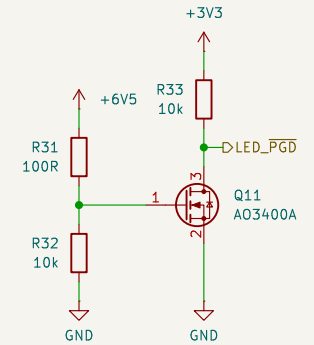
Id: 4/5

Power Supplies

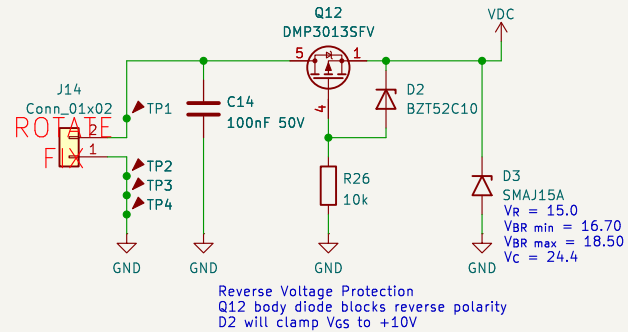
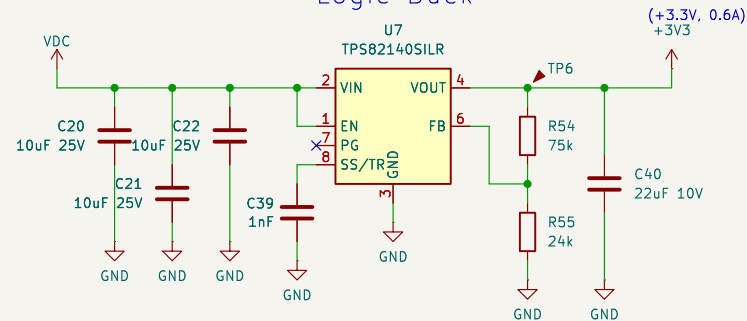
LED Buck



LED Power Good



Logic Buck



Find a good SMD Power input connector
lever terminal blocks?

EMC ferrites /CMC?

Wires & Bits

Sheet: /POWER/
File: power.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.9

Date: 2026-04-10

Rev: 0

Id: 5/5